Integration by Parts

ANSWER KEY

The integration by parts formula

1 Evaluate $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$: $= xe^x - \int e^x\,dx = xe^x - e^x + C$.

2 Evaluate $\int x\cos x\,dx$ using integration by parts.

Let $u = x$, $dv = \cos x\,dx$: $= x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

3 Evaluate $\int \ln x\,dx$ using integration by parts (with $dv = dx$).

Let $u = \ln x$, $dv = dx$: $= x\ln x - \int x \cdot \frac{1}{x}\,dx = x\ln x - x + C$.

Repeated and definite applications

4 Evaluate $\int x^2 e^x\,dx$, applying integration by parts twice.

First pass ($u = x^2$): $x^2e^x - 2\int xe^x\,dx$. Second pass gives $\int xe^x\,dx = xe^x - e^x$. Total: $x^2e^x - 2xe^x + 2e^x + C$.

5 Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.

Let $u = x$, $dv = e^{2x}dx$: $\left[\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x}\right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4}\right) - \left(0 - \frac{1}{4}\right) = \frac{e^2}{4} + \frac{1}{4}$.

Choosing u and dv wisely (LIATE)

6 Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).

$u = \ln x$, $dv = x\,dx$, so $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2}\ln x - \int \frac{x^2}{2} \cdot \frac{1}{x}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

7 Evaluate $\int x^2\cos x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = \cos x\,dx$: $x^2\sin x - 2\int x\sin x\,dx$. Second (on $\int x\sin x\,dx$): $= -x\cos x + \sin x$. Total: $x^2\sin x + 2x\cos x - 2\sin x + C$.

The recurring-integral trick

8 Evaluate $\int e^x\sin x\,dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

Let $I = \int e^x\sin x\,dx$. First pass ($u = \sin x$, $dv = e^x dx$): $I = e^x\sin x - \int e^x\cos x\,dx$. Second pass on the new integral ($u = \cos x$): $\int e^x\cos x\,dx = e^x\cos x + \int e^x\sin x\,dx = e^x\cos x + I$. So $I = e^x\sin x - e^x\cos x - I$, giving $2I = e^x(\sin x - \cos x)$, so $I = \frac{e^x(\sin x - \cos x)}{2} + C$.

Definite integrals by parts

9 Evaluate $\int_0^\pi x\sin x\,dx$ exactly.
$u=x$, $dv=\sin x\,dx$: $[-x\cos x+\sin x]_0^\pi=(-\pi\cos\pi+\sin\pi)-(0)=\pi$.

10 Evaluate $\int_1^e x^2\ln x\,dx$ exactly.
$u=\ln x$, $dv=x^2dx$: $\left[\frac{x^3}{3}\ln x-\frac{x^3}{9}\right]_1^e=\left(\frac{e^3}{3}-\frac{e^3}{9}\right)-\left(0-\frac{1}{9}\right)=\frac{2e^3}{9}+\frac{1}{9}=\frac{2e^3+1}{9}$.

Reduction formulas

11 Use integration by parts with $u=x^n$, $dv=e^xdx$ to derive the reduction formula $\int x^ne^x\,dx=x^ne^x-n\int x^{n-1}e^x\,dx$.
$du=nx^{n-1}dx$, $v=e^x$: $\int x^ne^x\,dx=x^ne^x-\int e^x\cdot nx^{n-1}\,dx=x^ne^x-n\int x^{n-1}e^x\,dx$.

12 Use the reduction formula above (applied twice) to evaluate $\int x^2e^x\,dx$, confirming it matches an earlier result.
$\int x^2e^x\,dx=x^2e^x-2\int xe^x\,dx$. Applying again: $\int xe^x\,dx=xe^x-\int e^x\,dx=xe^x-e^x$. Total: $x^2e^x-2(xe^x-e^x)=x^2e^x-2xe^x+2e^x+C$, matching the earlier double-integration-by-parts result.

Integration by parts with inverse trig functions

13 Evaluate $\int\tan^{-1}x\,dx$ using integration by parts with $u=\tan^{-1}x$, $dv=dx$.
$du=\frac{1}{1+x^2}dx$, $v=x$: $=x\tan^{-1}x-\int\frac{x}{1+x^2}\,dx=x\tan^{-1}x-\frac{1}{2}\ln(1+x^2)+C$.

14 Evaluate $\int x\sin^{-1}x\,dx$ using $u=\sin^{-1}x$, $dv=x\,dx$ (the resulting integral can be left in terms of a known antiderivative form).
$du=\frac{1}{\sqrt{1-x^2}}dx$, $v=\frac{x^2}{2}$: $=\frac{x^2}{2}\sin^{-1}x-\frac{1}{2}\int\frac{x^2}{\sqrt{1-x^2}}\,dx$, which requires a further trigonometric substitution to complete.

A capstone mixed-technique problem

15 Evaluate $\int_0^{\pi/2}x\cos(2x)\,dx$ exactly, using integration by parts.
$u=x$, $dv=\cos(2x)dx$, $v=\frac{1}{2}\sin(2x)$:
$\left[\frac{x}{2}\sin(2x)\right]_0^{\pi/2}-\frac{1}{2}\int_0^{\pi/2}\sin(2x)\,dx=\left(\frac{\pi}{4}\sin\pi-0\right)-\frac{1}{2}\left[-\frac{1}{2}\cos(2x)\right]_0^{\pi/2}=0-\frac{1}{2}\left(\frac{1}{2}+\frac{1}{2}\right)=-\frac{1}{2}$.

Visualizing an area found by integration by parts

16 The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.

$u = x$, $dv = e^{-x}dx$, $v = -e^{-x}$:
$[-xe^{-x}]_0^3 + \int_0^3 e^{-x}\,dx = -3e^{-3} + [-e^{-x}]_0^3 = -3e^{-3} + (1 - e^{-3}) = 1 - 4e^{-3} \approx 0.801.$

Choosing between substitution and parts

17 For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u-substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

(a) Substitution — $u = x^2$ makes $du = 2x\,dx$ appear directly, no product rule needed. (b) Integration by parts — x and e^{2x} are genuinely different function types with no simplifying substitution; use $u = x$, $dv = e^{2x}dx$.

18 Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

$du = \sec x \tan x\,dx$, $v = \tan x$:
$I = \sec x \tan x - \int \sec x \tan^2 x\,dx = \sec x \tan x - \int \sec x(\sec^2 x - 1)\,dx = \sec x \tan x - \int \sec^3 x\,dx + \int \sec x\,dx$. So $2I = \sec x \tan x + \ln|\sec x + \tan x|$, giving $I = \frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + C$.

19 A city's population growth rate is modeled by $r(t) = t\ln(t+1)$ thousand people per year, for $0 \le t \le 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.

Total growth $= \int_0^5 t\ln(t+1)\,dt$. This does not integrate cleanly with $u = \ln(t+1)$, $dv = t\,dt$ alone; using $u = \ln(t+1)$, $dv = t\,dt$, $v = \frac{t^2}{2}$: $\left[\frac{t^2}{2}\ln(t+1)\right]_0^5 - \frac{1}{2}\int_0^5 \frac{t^2}{t+1}\,dt$. Dividing $\frac{t^2}{t+1} = t - 1 + \frac{1}{t+1}$:
$\frac{1}{2}\int_0^5 \left(t - 1 + \frac{1}{t+1}\right)dt = \frac{1}{2}\left[\frac{t^2}{2} - t + \ln(t+1)\right]_0^5 = \frac{1}{2}(12.5 - 5 + \ln 6) \approx 4.65$. So total growth $\approx \frac{25}{2}\ln 6 - 4.65 \approx 22.41 - 4.65 \approx 17.76$ thousand people.